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Research Article

Gentian *FLOWERING LOCUS T* orthologs regulate phase transitions: floral induction and endodormancy release

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Abstract

Perennial plants undergo a dormant period in addition to the growth and flowering phases that are commonly observed in annuals and perennials. Consequently, the regulation of these phase transitions in perennials is believed to be complicated. Previous studies have proposed that orthologs of *FLOWERING LOCUS T* (*FT*) regulate not only floral initiation but also dormancy. We, therefore, investigated the involvement of *FT* orthologs (*GtFT1* and *GtFT2*) during the phase transitions of the herbaceous perennial gentian (*Gentiana triflora*). Analysis of seasonal fluctuations in the expression of these genes revealed that *GtFT1* expression increased prior to budbreak and flowering, whereas *GtFT2* expression was induced by chilling temperatures with the highest expression occurring when endodormancy was released. The expression of *FT*-related transcription factors, reportedly involved in flowering, also fluctuated during each phase transition. These results suggested the involvement of *GtFT1* in budbreak and floral induction and *GtFT2* in dormancy regulation, implying that the two gentian *FT* orthologs activated a different set of transcription factors. Gentian *ft2* mutants generated by CRISPR/Cas9-mediated genome editing had a lower frequency of budbreak and budbreak delay in overwintering buds caused by an incomplete endodormancy release. Our results highlighted that the gentian orthologs of *FRUITFULL* (*GtFUL*) and *SHORT VEGETATIVE PHASE-like 1* (*GtSVP-L1*) act downstream of *GtFT2*, probably to prevent untimely budbreak during ecodormancy. These results suggest that each gentian *FT* ortholog regulates a different phase transition by having variable responses to endogenous or environmental cues, leading to their ability to induce the expression of distinct downstream genes.

Introduction

Perennial plants can survive for several years by repeating the cycle of growth, flowering, and dormancy phases. By tightly regulating the transition and maintenance of these phases, perennials can grow, develop, and adapt to the different environmental conditions encountered over an entire year. Among the phases, dormancy is one of the most important features of the perennial lifestyle, e.g. bud dormancy occurs in most perennials to withstand harsh environmental conditions.

Bud dormancy is classified into the three distinct phases of paradormancy, endodormancy, and ecodormancy, each

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phase is affected by growth arrests in distal organs, internal bud signals, and environmental conditions, respectively (Lang, 1987). In Japanese gentians (*Gentiana triflora*, *G. scabra*, and their hybrids), overwintering buds (OWBs) are produced from underground tissues prior to winter and subsequently enter endodormancy. During winter, the phase transition from endodormancy to ecodormancy occurs; in the spring, OWBs initiate budbreak (Takahashi et al., 2014). After budbreak, Japanese gentians stay in the vegetative growth phase for several months, followed by a transition to the reproductive growth phase (Imamura et al., 2011).

Importantly, a shift from endodormancy to ecodormancy (endodormancy release) is necessary for budbreak in response to favorable conditions. Previous studies demonstrated that the bud dormancy phases are regulated by daylength, temperature, and phytohormones (Maurya and Bhalerao, 2017; Liu and Sherif, 2019). For example, short days induce growth cessation and bud set in hybrid aspen, whereas extended exposure to chilling induces endodormancy release in Japanese pear flower buds in conjunction with characteristic genetic or metabolic fluctuations (Böhlenius et al., 2006; Horikoshi et al., 2018). Among the phytohormones that play a role in bud dormancy, abscisic acid (ABA) is essential for establishing and maintaining endodormancy (Tylewicz et al., 2018; Singh et al., 2019). Other phytohormones such as gibberellins, auxins, cytokinins, and ethylene also serve as signal molecules regulating the dormancy phase transitions (Liu and Sherif, 2019). Furthermore, sugars are also involved in bud dormancy transitioning. Changes in sugar concentrations affect the expression of cell-cycle genes, and the progression or arrest of the cell cycle appears to be associated with dormancy phases (Horvath et al., 2002; Espinosa-Ruiz et al., 2004). In Japanese gentian OWBs, gentiobiose [β-D-Glc*p*-(1→6)-D-Glc] functions as a signal for budbreak through redox regulation. The concentration of gentiobiose significantly decreases from the endodormancy to ecodormancy stages, probably to prevent unfavorable budbreak during ecodormancy (Takahashi et al., 2014).

Notably, some orthologs of *FLOWERING LOCUS T* (*FT*) in perennial species are reported to be involved in regulating bud dormancy (reviewed in Singh et al., 2017). Generally, *FT* plays a central role in regulating floral initiation, which refers to the transition from the vegetative growth phase to the reproductive growth phase by integrating signals from multiple pathways to promote flowering (Teotia and Tang, 2015). In the annual Arabidopsis (*Arabidopsis thaliana*), *CONSTANS* (*CO*) expression is regulated by the plant's circadian clock and daylength, and the accumulated CO protein stimulates the expression of *FT* (Valverde, 2011). In contrast, temperature-dependent signals affect the expression of *FLOWERING LOCUS C* (*FLC*) and *SHORT VEGETATIVE PHASE* (*SVP*), thereby suppressing the expression of two floral integrator genes, *FT* and *SUPPRESSOR OF OVEREXPRESSION OF CO1* (*SOC1*; Helliwell et al., 2006; Lee and Lee, 2010). Other reports have shown that *SOC1* expression is positively regulated not only by *FT* but also by *SQUAMOSA PROMOTER BINDING PROTEIN-LIKEs* (*SPLs*) in response to plant age and phytohormones (Lee and Lee, 2010; Yu et al., 2012). Upregulated expression of *FT*, *SOC1*, and *SPLs* combined with the downregulated expression of the floral repressor *TERMINAL FLOWER1* (*TFL1*) and the activation of floral meristem identity genes, including *APETALA 1* (*AP1*), *FRUITFULL* (*FUL*), and *LEAFY* (*LFY*), initiates flower development (Hanano and Goto, 2011; Teotia and Tang, 2015). Although the molecular basis for floral induction and development centers on *FT* and is conserved in various perennials, *FT* also regulates growth cessation, bud set, and dormancy (Maurya and Bhalerao, 2017; Singh et al., 2018). Poplar (*Populus* spp.) orthologs of *FT*, *PtFT1*, and *PtFT2* have different functions and are regulated by other environmental cues (Hsu et al., 2011). Upregulated by low temperature, *PtFT1* promotes floral initiation but can also prevent short-day-induced dormancy in flowering buds. In contrast, *PtFT2* is upregulated by warm temperatures, and long-day photoperiods during vegetative growth inhibit bud set. Furthermore, several reports suggest that *FT* orthologs are expressed in parallel with cell cycle-, phytohormone-, and sugar-related genes to regulate dormancy phase transitions (Rinne et al., 2011; Vergara et al., 2016; Zhang et al., 2018). Although these observations indicate the possibility that *FT* orthologs have multiple functions or the orthologs were functionally differentiated to regulate dormancy phase transitions, these hypotheses remain to be fully confirmed.

This study investigated whether *FT* orthologs have diverse functions during the several phase transitions in the herbaceous perennial *Gentiana*. Our previous study demonstrated that gentian *FT* orthologs (*GtFT1* and *GtFT2*) are associated with flowering, i.e. the expression levels of *GtFT1* increased prior to floral initiation; high-levels and early expression were observed in early-flowering varieties, and the overexpression of *GtFT1* hastened flowering in gentian plantlets and Arabidopsis (Imamura et al., 2011). It was questionable, however, whether *GtFT2* mainly functioned as a flowering inducer because the timing for *GtFT2* expression did not correspond with the flowering period for field-grown Japanese gentians. Here, we obtained direct evidence that *GtFT2* regulates endodormancy release, rather than floral induction, using CRISPR/Cas9 technology. Our results also support the hypothesis that *GtFT1* and *GtFT2* act with different sets of transcription factors and regulate budbreak, flowering, and dormancy. We propose that *FT* orthologs in perennials can be functionally differentiated and may be involved individually in different phase transitions during a perennial plant's life cycle.

Results

Seasonal expression of *FT* orthologs in field-grown gentians

Gentians have two dormancy phases, i.e. endodormancy (October to November) and ecodormancy (December to March), and two growth phases, i.e. vegetative growth (April

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to May) and reproductive growth (June or July to September; Imamura et al., 2011; Takahashi et al., 2014). In this study, we characterized the function of gentian *FT* orthologs during phase transitions other than floral initiation. First, we identified nucleotide sequences of *GtFT1* and *GtFT2* in *G. triflora* cv. SpB and obtained sequences that were perfectly matched to those in *G. triflora* shown in Imamura et al. (2011). Based on these sequences, we measured the seasonal fluctuation of *GtFT1* and *GtFT2* expression in leaves and shoot apical meristems (SAMs) of field-grown gentians (*G. triflora* cv. SpB) throughout 1 year (Figure 1A). The relative expression level of *GtFT1* was low during dormancy but began to increase in leaves before the onset of budbreak (Figure 1A, inset) and continued to increase until flowering. The transcript abundance of *GtFT1* in SAMs was lower than that of leaves, although a slight increase was observed from April to September. In contrast, *GtFT2* in leaves and SAMs was expressed at its highest levels in December when the dormancy phase in OWBs transits from endodormancy to ecodormancy, with another slight increase in August. Therefore, it appeared that *GtFT1* expression was associated with budbreak and flowering, but *GtFT2* expression was associated with endodormancy release. Given that the highest expression of *GtFT2* was observed during endodormancy release, we also investigated the expression of *GtFT2* and *GsFT2* using the OWBs of *G. triflora* cv. Iwate and *G. scabra* cv. Alta, which are early-flowering and late-flowering varieties, respectively (Figure 1B). These varieties showed a similar expression pattern to that observed in *G. triflora* cv. SpB, suggesting that the observed *GtFT2* expression during dormancy appeared to be a common feature of gentians.

Furthermore, we found that the expression of *GtFT1* and *GtFT2* had different responses to being chilled (Figure 2). We previously reported that chilling induces endodormancy release, i.e. gentian OWBs enter endodormancy after zero to 3 weeks incubation at 4°C but enter ecodormancy 5 weeks after incubation at 4°C (Takahashi et al., 2014). In the OWBs of *G. triflora* cv. SpB, the expression level of *GtFT1* was unchanged during chilling, whereas the expression level of *GtFT2* increased in response to chilling for 3 weeks but decreased 5 weeks after chilling when the dormancy phase transitioned to ecodormancy. These results indicated that *GtFT2* had the highest expression level prior to endodormancy release.

Generation of *ft2* mutants in gentians using the CRISPR/Cas9 system

To confirm the role of gentian *FT2* during dormancy, we produced gentian *ft2* mutants in cv. Albireo by CRISPR/Cas9-mediated genome editing (Figure 3). Since cv. Albireo is a hybrid cultivar and possesses two alleles derived from *G. triflora* and *G. scabra*, namely *Alb-GtFT2* and *Alb-GsFT2*, a 20-bp sequence in the first exon of *FT2* that is identical in both alleles was selected as a target sequence (Figure 3A; Supplemental Figure S1). We obtained 53 transgenic

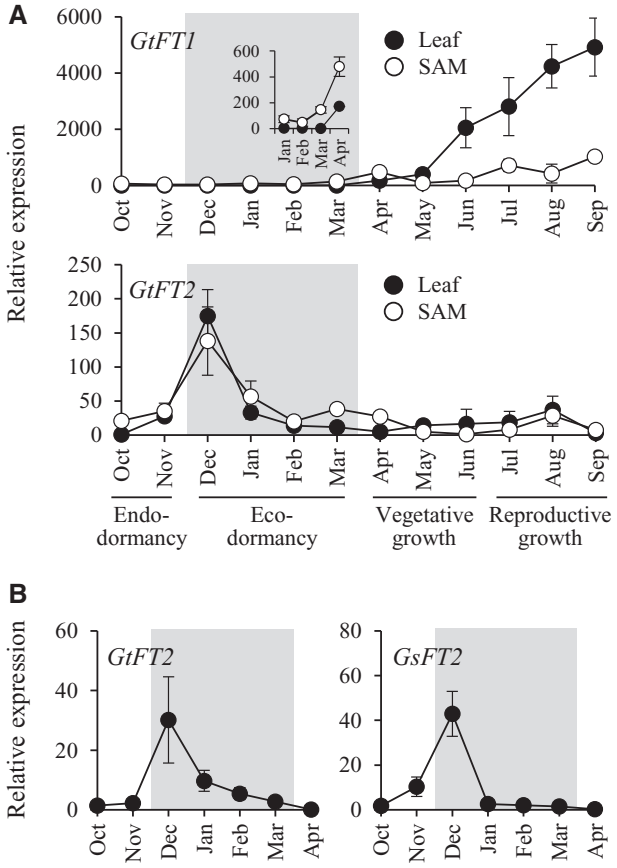


Figure 1 Expression of *FT* genes in field-grown gentian plants throughout all stages of dormancy and growth. A, Seasonal expression of *GtFT1* and *GtFT2* in leaves (filled circles) and SAM (open circles) in *G. triflora* cv. SpB. The inset shows an amplified plot of the expression from January to April. B, Expression of *FT2* genes in OWBs of *G. triflora* cv. Iwate (*GtFT2*; left) and *G. scabra* cv. Alta (*GsFT2*; right). The ecodormancy period is represented by a shaded area. The relative expression level of each gene is normalized to that of *Actin*, with expression in leaves in October defined as 1. Each value represents the means ± SD calculated from five independent biological replicates. Five biological replicates were prepared each being a single tissue (SAM or leaf) from a plant.

plantlets and confirmed that the mutations were generated at the target site. After sequencing the genomic region and cDNA corresponding to the *FT2* locus, we selected two frame-shift mutants, denoted as *ft2#35* and *ft2#53*. The *ft2#35* mutant had 7- and 5-nucleotide (nt) deletions in the *Alb-Gtft2*$^{del129-135}$ and *Alb-Gsft2*$^{del130-134}$ alleles, respectively. The *ft2#53* mutant had 4- and 5-nt deletions in *Alb-Gtft2*$^{del131-134}$ and *Alb-Gsft2*$^{del130-135}$ alleles, respectively. Since these mutations generated premature stop codons predicted to result in truncated FT2 proteins (Figure 3B), we used these mutants in further analyses. First, we investigated whether the *ft2* mutations influence plantlet phenotypes and found that the *ft2* mutants had similar morphologies and growth characteristics as the wild-type (WT; Supplemental Figure S2A). We also compared soil-grown WT and *ft2* mutant plants but found almost no difference in their shoot growth and developmental characteristics;

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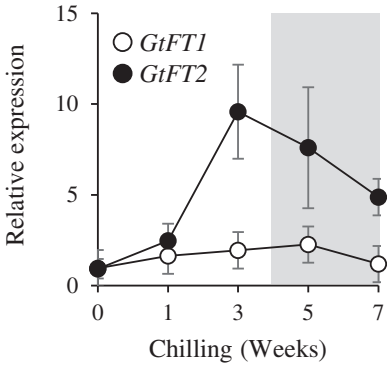


Figure 2 Expression of *GtFT1* and *GtFT2* in chilled OWBs. Field-grown OWBs of cv. SpB were harvested in October and chilled at 4°C for zero to 7 weeks. The shaded area represents the ecodormancy phase shown by Takahashi et al. (2014). The relative expression level is normalized to *Actin* and expression for week zero was defined as 1. Each value represents the mean ± SD calculated from three independent biological replicates. Three biological replicates were prepared each being a single OWB tissue from a plant.

e.g. the timing of floral initiation was normal in the mutants (Supplemental Figure S2B).

Mutations in *FT2* affect budbreak but do not affect OWB formation

We also compared the morphology of OWBs and the rhizospheres of the WT and *ft2* mutants (Figure 4A). The appearance of OWBs and rhizospheres was nearly identical, and there was no significant difference in the number of OWBs per plant (13.7 ± 1.5 in the WT, 13.3 ± 2.5 in *ft2#35*, and 12.7 ± 3.8 in *ft2#53*). The OWBs of the WT and *ft2* mutants were chilled at 4°C for 7 weeks before being transferred to an optimal growing temperature of 20°C to induce budbreak. Three weeks after transfer, the OWBs of the WT had already sprouted and shoots began to grow, but those of *ft2* mutants had just started budbreak (Figure 4B). Five weeks after the transfer, there was a noticeable difference in shoot growth between the WT and the *ft2* mutants (Figure 4C). Furthermore, we compared the budbreak frequency and shoot length between WT and *ft2* mutants. Although approximately 60% of the OWBs sprouted in the WT by 3 weeks after transfer, only 24% and 30% of the OWBs sprouted in the *ft2#35* and *ft2#53* mutants, respectively (Figure 4D). Five weeks after transfer, the budbreak frequency increased slightly in the WT and the *ft2* mutants compared to 3 weeks after transfer, but the level in the *ft2* mutants was significantly lower than that of the WT (Figure 4E). The budbreak frequency did not increase any further, suggesting that the *ft2* mutation affected the timing of budbreak and also influenced the budbreak frequency of the OWBs. A comparison of shoot growth after budbreak revealed that the shoot length of the *ft2#35* mutant tended to be shorter than the WT, whereas shoot length in the *ft2#53* mutant was significantly shorter than the WT (Figure 4F).

The dormancy phase transition in gentian OWBs is affected by *FT2*

Because endodormancy release is necessary for budbreak, we hypothesized that the *FT2* mutation suppressed the release, resulting in the delay and suppression of budbreak. In gentian OWBs, the gentiobiose concentration decreased significantly after endodormancy release, suggesting that gentiobiose can be used as a marker to distinguish between the endodormancy and ecodormancy phases (Takahashi et al., 2014). To investigate the effect of *ft2* mutations on endodormancy release, we measured the gentiobiose concentrations in OWBs of the WT and *ft2* mutants during endo- and ecodormancy (Figure 5). During endodormancy, no significant difference was observed in the gentiobiose concentration between the WT and *ft2* mutants. Five weeks after chilling, the dormancy phase had transitioned from endodormancy to ecodormancy, and the gentiobiose concentration decreased by > 70% in WT plants. In the *ft2* mutants, the decrease in the gentiobiose concentration from 3 to 5 weeks after chilling was negligible, suggesting that endodormancy release was incomplete in the OWBs of *ft2* mutants. We also compared the sucrose concentration in the WT and *ft2* mutants because sucrose accumulation during chilling treatments has been reported in other plant species (Tabaei-Aghdaei et al., 2003). The sucrose concentration tended to increase with chilling, but there were no significant differences in sucrose content between WT plants and *ft2* mutants. The one exception was that a lower sucrose concentration was observed in the *ft2#53* mutant after chilling for 7 weeks, suggesting that chilling sensitivity was unchanged in this mutant.

The *ft2* mutation affects the expression of flowering-related genes in gentian OWBs

We found that the frame-shift mutation in *FT2* was caused by CRISPR/Cas9, resulting in incomplete endodormancy release and suppressed budbreak; however, the mechanism by which *FT2* regulates this process remained unclear. Pérez et al. (2011) previously demonstrated that flowering-related genes in grapevine (*Vitis vinifera*) buds helped regulate the induction and maintenance of endodormancy with *VvFT*. We, therefore, investigated the expression of gentian orthologs of genes reported to be involved in flowering regulation. A search of a gentian EST library produced the candidate orthologs, *GtFLC-L*, *GtSOC1a*, *GtSOC1b*, *GtFUL*, *GtSPL4*, *GtAP1*, and *GtLFY* (Figure 6A). Alignments of the deduced amino acid sequences of the seven orthologs are shown in Supplemental Figure S3. Sequence information for *GtSVP-L1* was provided from a previous study (Yamagishi et al., 2016). First, we determined the expression levels of flowering-related genes in the OWBs of the WT and *ft2* mutants treated at 4°C for 3 or 7 weeks; for the WT, the dormancy phases at 3 and 7 weeks corresponded to endodormancy or ecodormancy, respectively (Figure 6B). Fluctuations in gene expression occurred when the dormancy phase changed; however, there were no significant difference in the expression of *GtFLC-L*, *GtSOC1a*, *GtSPL4*, *GtAP1*, and *GtLFY* between the WT and *ft2* mutants. Although the expression level of *GtSOC1b* in endodormant OWBs was similar to

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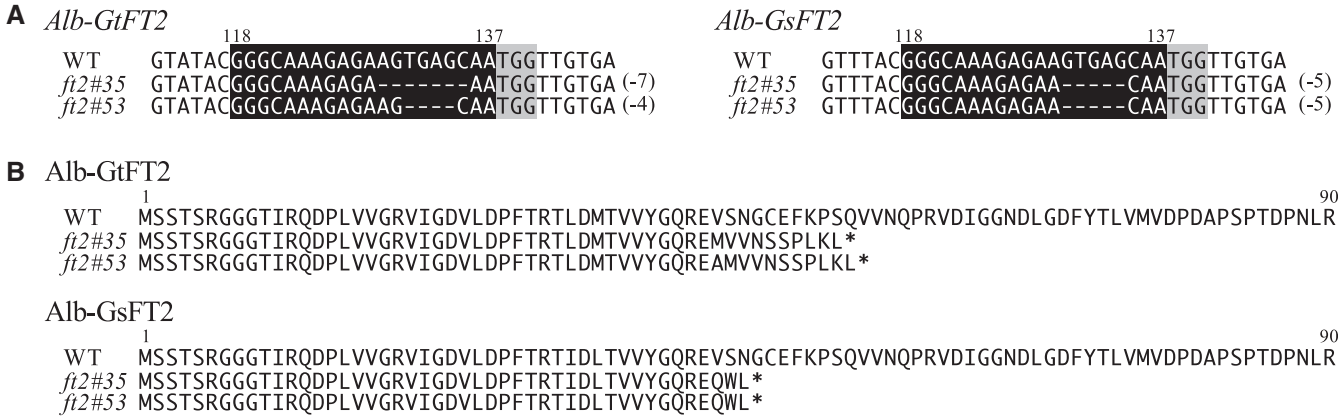


Figure 3 CRISPR/Cas9-mediated mutations in the gentian *FT2* gene in a hybrid gentian, *G. triflora* (Gt) × *G. scabra* (Gs). A, Nucleotide sequences derived from *GtFT2* and *GsFT2* alleles of the WT and mutants with small deletions in the *FT2* gene (*ft2#35* and *ft2#53*). The sgRNA target sequences are shown as white text on a black background and protospacer adjacent motif sequences are highlighted in gray. Deletions are indicated by dashes and the numbers in parentheses on the right side indicate the size of the deletions. B, Deduced amino acid sequences of the WT and *ft2* mutants. Asterisks indicate the position of newly generated stop codons.

that in ecodormant OWBs in all lines, one of the *ft2* mutants had a lower expression level of *GtSOC1b* compared to the WT. Importantly, *GtFUL* and *GtSVP-L1* expression levels increased during the transition from endodormancy to ecodormancy in the WT, but the increase in expression was suppressed significantly in the *ft2* mutants.

Seasonal expression of the gentian orthologs of flowering-related genes was also investigated in cv. SpB (Figure 6C). *GtFLC-L* expression in leaves and SAMs decreased from November to February and remained low after budbreak. The transcript level of *GtSOC1a* in leaves and SAMs tended to decrease from October to September, except for a transient increase in SAMs observed in April. The expression of *GtSOC1b* in leaves increased gradually, with the highest expression levels observed in July and a transient increase observed in SAMs in April. Interestingly, the expression levels of *GtFUL* increased 3.5- and 8.9-fold in leaves and SAMs, respectively, during the transition from endodormancy (November) to ecodormancy (December). In contrast, the expression level of *GtFUL* decreased when budbreak occurred in April. In leaves, *GtFUL* expression increased again in June but decreased in July. The transcript abundance of *GtSPL4* and *GtAP1* showed a similar trend, with expression levels in SAMs being low from endodormancy to the vegetative growth phase before increasing transiently in July when reproductive growth began. *GtSPL4* expression in leaves increased slightly towards the onset of the flowering period, whereas *GtAP1* expression increased transiently in August. Transcript abundance of *GtLFY* in SAMs increased during ecodormancy before decreasing in April and increasing again in June. In contrast, *GtLFY* expression in leaves was low during dormancy, before increasing gradually from March to August and then decreasing in September. The expression of *GtSVP-L1* in SAMs mirrored the expression of *GtLFY* in SAMs; however, the expression of *GtSVP-L1* in leaves was relatively higher during ecodormancy compared to that during other growth phases.

Mutation of *FUL* inhibits dormancy of gentian OWBs

Since the expression of *GtFUL* increased during ecodormancy and *ft2* mutations significantly affected *GtFUL* expression, we generated *ful* mutants by CRISPR/Cas9 technology to examine the relationship between *GtFUL* and ecodormancy (Figure 7). We constructed a binary CRISPR/Cas9 vector containing tandem single-guide RNAs (sgRNAs) that targeted the first exon (target sequence 1) and the second exon (target sequence 2) of *FUL*, respectively (Supplemental Figure S4A). We obtained 27 transgenic plants of *G. triflora* x *G. scabra* cv. Albireo and selected two frame-shift mutants (*ful#25* and *ful#27*; Figure 7A; Supplemental Figure S4, B and C). DNA sequence analysis revealed that *ful#25* had 14- and 10-nt deletions in target sequence 1 of the *Alb-Gtful*$^{del35-48}$ and *Alb-Gsful*$^{del41-50}$ alleles, respectively, and 1-nt insertion and 3-nt deletions in target sequence 2 of the *Alb-Gtful*^{in235} and *Alb-Gsful*$^{del231-233}$ alleles, respectively. The *ful#27* mutant had 4- and 14-nt deletions in target sequence 1 of the *Alb-Gtful*$^{del42-45}$ and *Alb-Gsful*$^{del32-45}$ alleles, respectively, and 2- and 4-nt deletions in target sequence 2 of the *Alb-Gtful*$^{del230-231}$ and *Alb-Gsful*$^{del230-233}$ alleles, respectively. These mutations generated premature stop codons that were predicted to result in truncated GtFUL and GsFUL proteins (Supplemental Figure S4B). We found that vegetative growth was inhibited in both *ful* mutants; in particular, stem growth and flowering were strongly inhibited (Figure 7B). Furthermore, a comparison of OWB appearance revealed that the number and size of OWBs were lower in the *ful* mutants than in the WT (Figure 7C). Most importantly, the OWBs of the *ful* mutants had already sprouted when the OWBs of the WT were still in the endodormancy phase. These results suggested that *GtFUL* had multiple functions in flowering and dormancy, although the detailed mechanisms underlying these are currently unknown (Figure 8).

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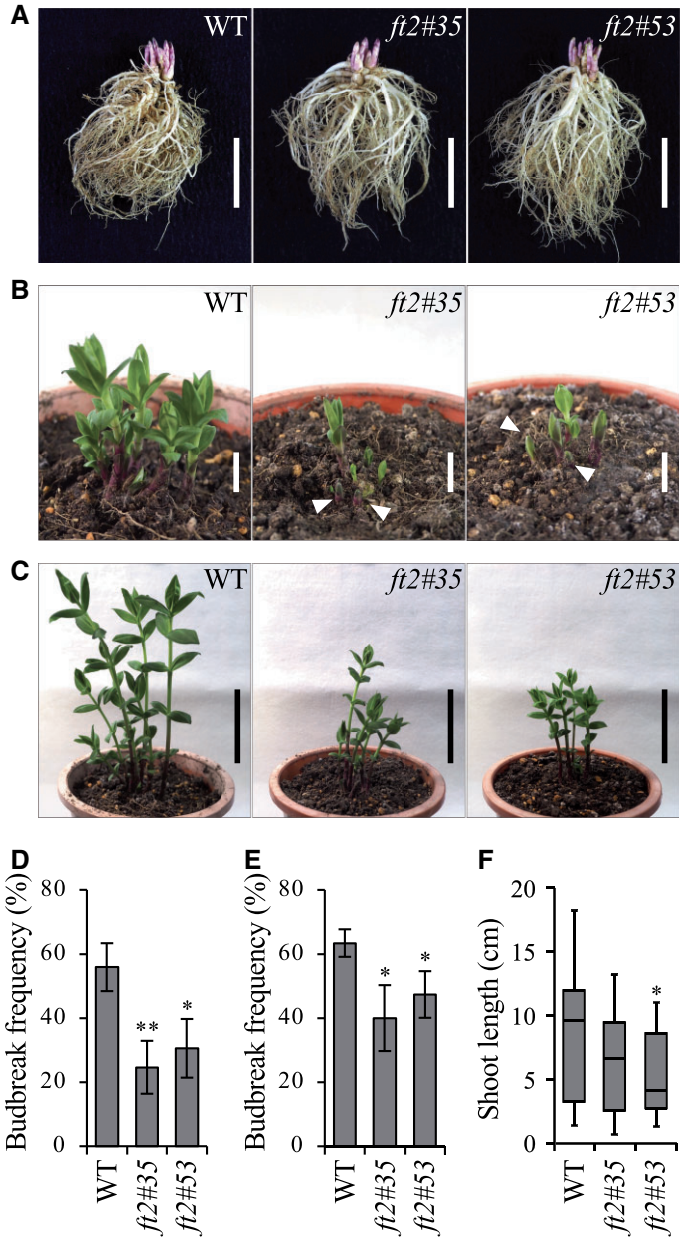


Figure 4 Suppression of budbreak and growth in OWBs by the *ft2* mutation. After a chilling treatment for 7 weeks, OWBs were transferred to an optimal growth temperature (20°C) and budbreak and growth were measured. A, Morphology of OWBs and underground organs in the WT and *ft2* mutants (*ft2#35* and *ft2#53*). Scale bars indicate 5 cm. B, Budbreak of OWBs occurred 3 weeks after transfer. Arrowheads indicate unsprouted buds. Scale bars indicate 1 cm. C, Shoots of WT and *ft2* mutants at 5 weeks after transfer. Scale bars indicate 5 cm. D, Budbreak frequency of WT ($n = 41$), *ft2#35* ($n = 40$), and *ft2#53* ($n = 38$) at 3 weeks after transfer. E, Budbreak frequency of WT ($n = 41$), *ft2#35* ($n = 40$), and *ft2#53* ($n = 38$) at 5 weeks after transfer. Each value represents the mean ± SD. F, Box and whisker plot of shoot length in the WT ($n = 26$), *ft2#35* ($n = 16$), and *ft2#53* ($n = 18$) at 5 weeks after transfer. The top and bottom of the box represent the upper and lower quartile, respectively, and lines inside box represent the median. The whiskers represent the highest and lowest values. Significant differences from WT are shown ($^{*}P < 0.05$, $^{**}P < 0.01$, Student's *t* test).

Discussion

Possible functional differentiation of gentian *FT* orthologs

Previously, we showed that the timing of *GtFT1* expression matched the flowering time of field-grown gentian plants (Imamura et al., 2011). That is, *GtFT1* expression began to increase approximately 1 month before floral initiation, and the overexpression of *GtFT1* in gentian plantlets or Arabidopsis induced early flowering. These results indicated that *GtFT1* might act as a flowering promoter; however, we demonstrated here that the increase in *GtFT1* expression started prior to budbreak in OWBs. The relationship between *FT* orthologs and budbreak has been reported in several perennial species (Kotoda et al., 2010; Niu et al., 2015; Singh et al., 2018). Additionally, Varkonyi-Gasic et al. (2013)

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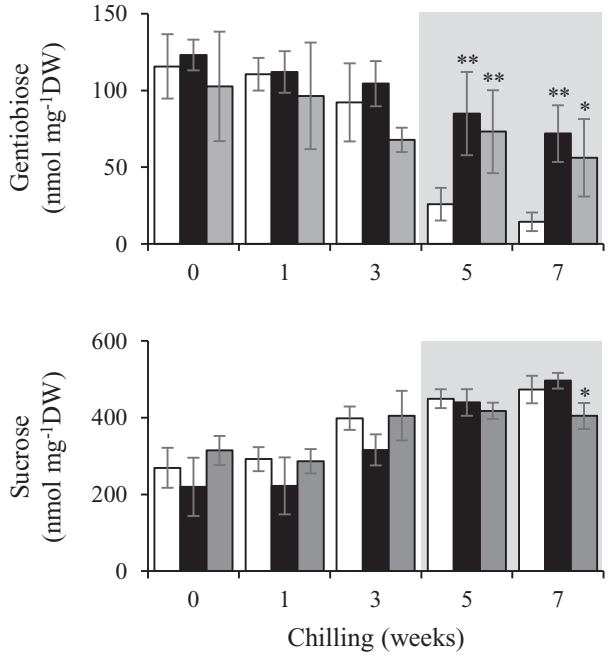


Figure 5 Effect of *FT2* impairment on gentiobiose and sucrose concentrations in gentian OWBs during endo- and ecodormancy. The concentrations in the WT (white bars), *ft2#35* (black bars), and *ft2#53* (gray bars) were compared. The predicted ecodormancy phase is represented by the shaded area. Each value represents the mean ± SD calculated from five independent biological replicates. Five biological replicates were prepared each being a single OWB tissue from a plant. Significant differences from the WT are shown ($^{*}P < 0.05$, $^{**}P < 0.01$, Student's *t* test).

revealed that *FT* expression began to increase before budbreak in kiwifruit, suggesting that the increased expression of *FT* promotes budbreak and subsequent flowering. This observation is noteworthy since Japanese gentian species are in the vegetative growth phase for over 2 months after budbreak, and budbreak can be clearly distinguished from flowering. *GtFT1* expression in gentian increased in two stages: one stage occurred prior to budbreak and the other stage was a further increase prior to floral initiation. Therefore, *GtFT1* possibly is multifunctional in inducing budbreak and floral initiation.

In contrast to *GtFT1*, the highest expression of *GtFT2* was observed when endodormancy release occurred. There was no effect of daylength on *GtFT1* and *GtFT2* expressions, and it is also not known as to which environmental factors affect the expression of *GtFT1* and *GtFT2* (Imamura et al., 2011). We found here that chilled OWBs induced *GtFT2* expression, but did not affect *GtFT1* expression. The difference in inducing factors between *GtFT1* and *GtFT2* agrees with a previous finding (Hsu et al., 2011). Chilling-inducible *FT* orthologs involved in endodormancy release have been identified in perennials, such as kiwifruit (*Actinidia* spp.) and poplar (Rinne et al., 2011; Varkonyi-Gasic et al., 2013; Singh et al., 2018). In perennials, *DORMANCY-ASSOCIATED MADS-BOX* (*DAM*) is well studied among the *FT*-related genes that regulate bud dormancy (Yamane et al., 2021). Since *DAM* genes are upregulated during dormancy induction and maintenance and then negatively regulate the expression of *FT* orthologs, *DAMs* are one of the main factors responsible for regulating dormancy in several perennial species (Falavigna et al., 2019). We did not find any *DAM* orthologs from the gentian expressed sequence tag (EST) library that was constructed by using multiple organs from various developmental stages. The possibility remains that the orthologs were expressed in other periods or organs that were not used for EST library construction. Although further investigations are still needed, it is possible that chilling-induced suppression of the expression of *DAM* orthologs or genes with similar function is involved in the upregulation of *GtFT2*.

Importantly, we previously revealed that the overexpression of *GtFT2* induced early flowering in gentian and Arabidopsis (Imamura et al., 2011), whereas *GtFT2* is also apparently involved in endodormancy release as observed in this study. These results indicate that gentian *FT* paralogs are capable of inducing flowering when host plants are in the vegetative growth phase; however, based on differences in the timing and location of their expression, the *FT* paralogs seem to have a variety of functions. In addition, we found candidate genes that may play roles in phase transitions under the control of *GtFT1*, i.e. *GtSOC1b* was expressed in SAMs prior to budbreak, but *GtLFY* and *GtAP1* were expressed in SAMs prior to floral initiation (Figure 6). Many reports have demonstrated that *FT* activates floral meristem identity genes such as *LFY* and *AP1* as well as the MADS transcription factor *SOC1* to promote flowering (reviewed in Teotia and Tang, 2015). This finding implies that *GtFT1* activates different transcription factors depending on the season. Although further studies are necessary to confirm these regulatory processes, our results highlight the possibility that *GtFT1* activates phase-specific sets of transcription factors to regulate the phase transitions suitable for the environment.

Gentian *FT2* accelerates endodormancy release and affects ecodormancy by controlling *GtFUL* and *GtSVP-L1* expression

In this study, we produced two *ft2* mutants (*ft2#35* and *ft2#53*) using the CRISPR/Cas9 system optimized for gentian (Tasaki et al., 2019) to obtain direct evidence that *GtFT2* affects endodormancy release. The mutation was predicted to lead to a truncated GtFT2 protein lacking approximately 130 amino acids in an essential region for evaluating FT function (Abe et al., 2005; Hanzawa et al., 2005). Consequently, the function of the *GtFT2* gene in these mutants was expected to be disrupted. We found that the *ft2* mutants showed no differences in growth or flowering time compared to the WT, indicating that the primary function of *FT2* is not floral induction, although it may play a supplementary role. Furthermore, since there were no differences in the appearance and number of OWBs or in the growth of underground organs of the *ft2* mutants compared to the WT, it appeared that *FT2* functions mainly after the initiation of endodormancy. Our results showed that the *ft2*

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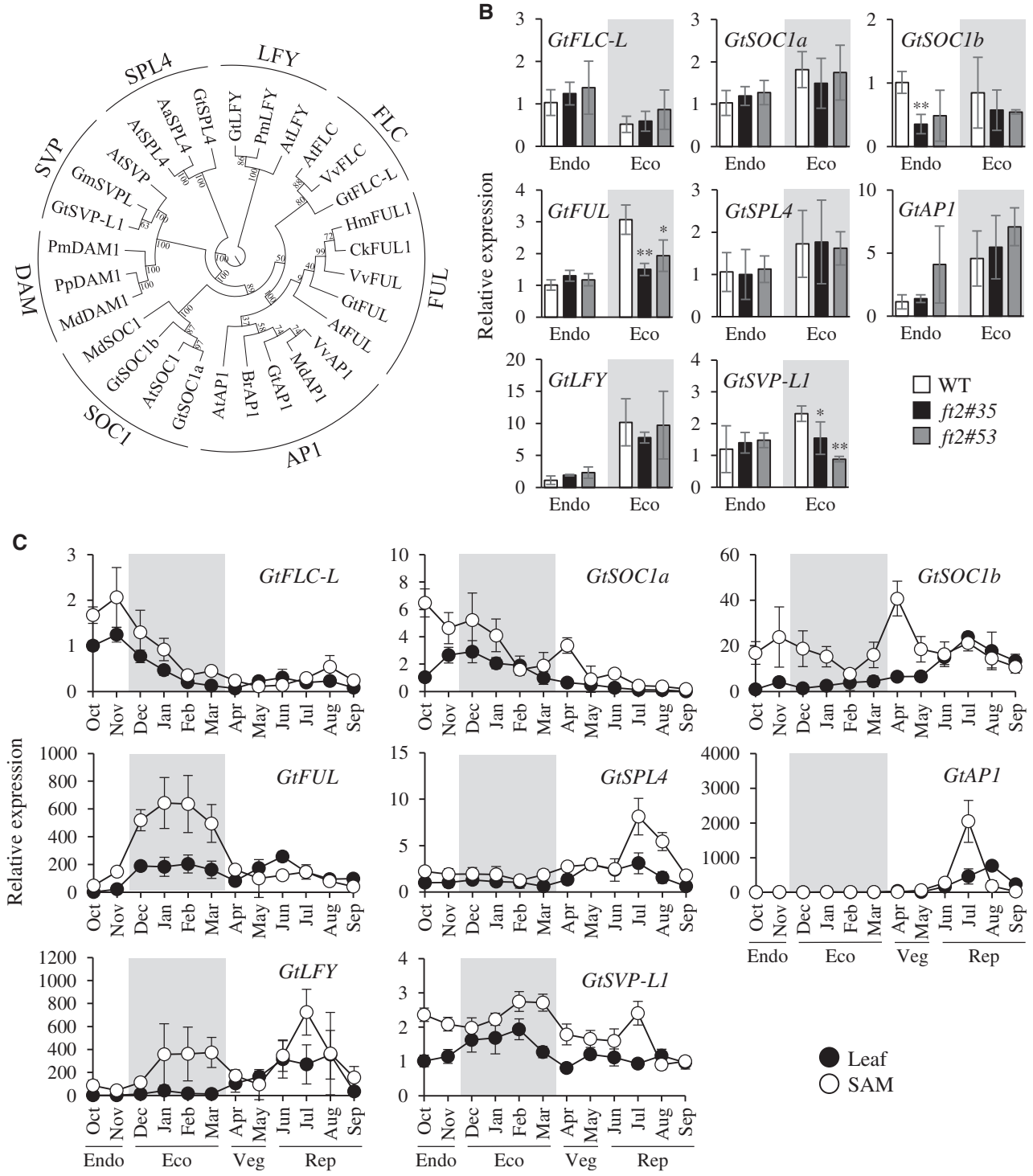


Figure 6 The *ft2* mutation affects the expression of flowering-related genes in gentians. A, A phylogenetic tree based on the deduced amino acid sequences of gentian orthologs of *FT*-related genes denoted as GtFLC-L, GtSOC1a, GtSOC1b, GtSVP-L1, GtFUL, GtSPL4, GtAP1, and GtLFY and their orthologs from *Arabidopsis thaliana* (AtFLC, Q9S7Q7; AtSOC1, O64645; AtSVP, Q9FVC1; AtFUL, Q38876; AtSPL4, Q9S7A9; AtAP1, P35631; AtLFY, Q00958), *Arabis alpina* (AaSPL4, S4WUK1), *Brassica rapa* (BrAP1, P0DI14), *Cornus kousa* (CkFUL1, M4IT41), *Glycine max* (GmSVPL, E3NYP9), *Hydrangea macrophylla* (HmFUL1, B5BUX3), *Malus domestica* (MdAP1, Q6YNE5; MdSOC1, Q09JE1; MdDAM1, A0A1B2LP93), *Prunus mume* (PmLFY, A0A140FAH6; PmDAM1, A0A2H4N2C0), *Pyrus pyrifolia* (PpDAM1, A0A0D4ZYJ8), and *Vitis vinifera* (VvFLC, D1MDP4; VvFUL, D7SMN6; VvAP1, Q6E6S7). Bootstrap values above the nodes were obtained from 1,000 replicates. B, A comparison of expression levels of flowering-related genes in gentian OWBs of the WT (white bars), *ft2#35* (black bars), and *ft2#53* (gray bars) during endodormancy and ecodormancy. The endodormant WT expression data were normalized to 1. C, Seasonal expression of flowering-related genes in leaves (solid circles) and SAMs (open circles) of gentians, covering endodormancy (Endo), ecodormancy (Eco), the vegetative growth phase (Veg), and the reproductive growth phase (Rep). The leaf expression data for October were normalized to 1. The predicted ecodormancy phase is represented by the shaded area. In (B) and (C), each value represents the mean ± SD calculated from five independent biological replicates. Five biological replicates were prepared each being a single tissue (OWB, leaf or SAM) from a plant. Significant differences from the WT are shown ($^{*}P < 0.05$, $^{**}P < 0.01$, Student's *t* test).

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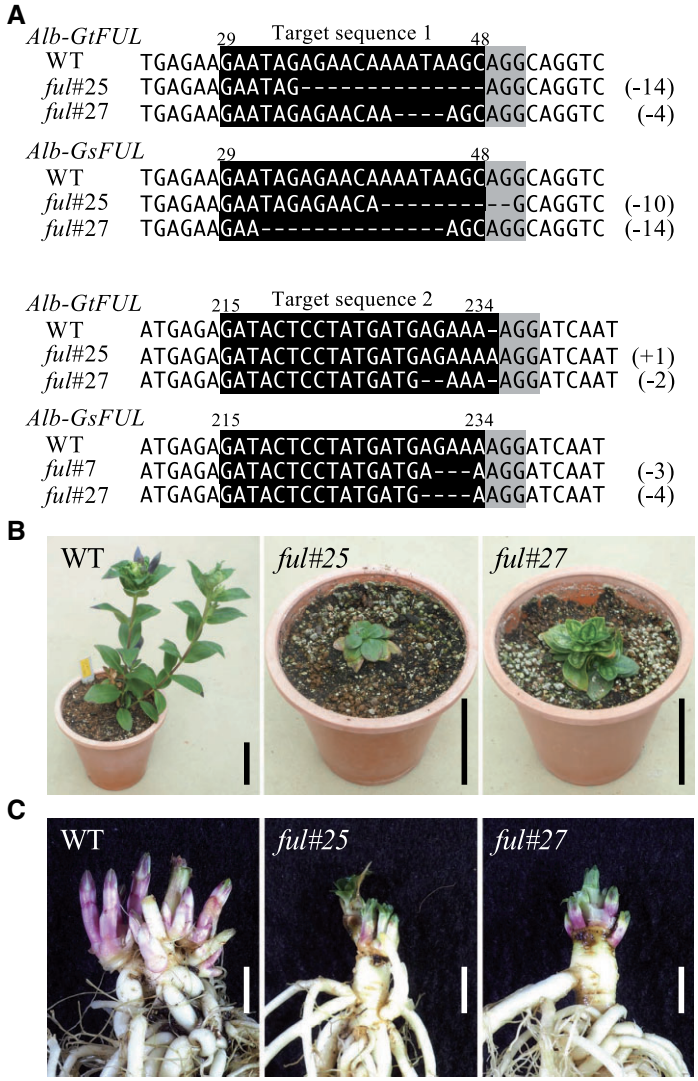


Figure 7 CRISPR/Cas9-mediated genome editing of *FUL* in gentian plants. A, The DNA sequence of *FUL* in WT, *ful#25*, and *ful#27* plantlets. The sgRNA target sequences are shown as white text on a black background and protospacer adjacent motif sequences are highlighted in gray. Deletions are indicated by dashes and the numbers in parentheses on the right side indicate the size of the deletions. B, Photographs of 4-month-old WT, *ful#25*, and *ful#27* plantlets. Scale bars indicate 5 cm. C, Comparison of phenotypes of OWBs of WT, *ful#25*, and *ful#27*. Scale bars indicate 1 cm.

mutations impaired budbreak in OWBs. Budbreak occurrence in the *ft2* mutants was lower than that of WT plants at both 3 and 5 weeks after being transferred to optimal growth temperatures. These findings indicate that some of the OWBs in *ft2* mutants lost the ability to sprout, and, even among those that did sprout, budbreak was delayed. Consequently, we hypothesize that an abnormal dormancy transition occurred in the OWBs of the *ft2* mutants. In gentian OWBs, dormancy phases are easily distinguished because the gentiobiose concentration decreases markedly after endodormancy release (Takahashi et al., 2014). Despite the apparent decrease in gentiobiose concentration in the OWBs of WT plants, our experiments documented that the decrease was significantly suppressed in the OWBs of the *ft2* mutants. This result suggests that endodormancy release was not completed successfully in the *ft2* mutants. Our results also showed that the gentiobiose concentration decreased gradually in the chilled *ft2* mutants, implying that the process of endodormancy release was not arrested but proceeded slowly. These results collectively indicate that *GtFT2* functions as an accelerator for endodormancy release in gentian OWBs, and that the delay in endodormancy release due to the *GtFT2* mutation resulted in delayed budbreak.

We also found that the *ft2* mutations repressed the transcript abundance of *GtFUL* and *GtSVP-L1*, indicating that endodormancy release is necessary to increase the expression levels of *GtFUL* and *GtSVP-L1*. Seasonal expression analysis in field-grown gentian plants revealed that the expression levels of *GtFUL* and *GtSVP-L1* increased during endodormancy and were further elevated after endodormancy release but decreased concurrently with the onset of budbreak. Additionally, OWBs of the *ful* mutants exhibited budbreak even though the dormancy phase did not change

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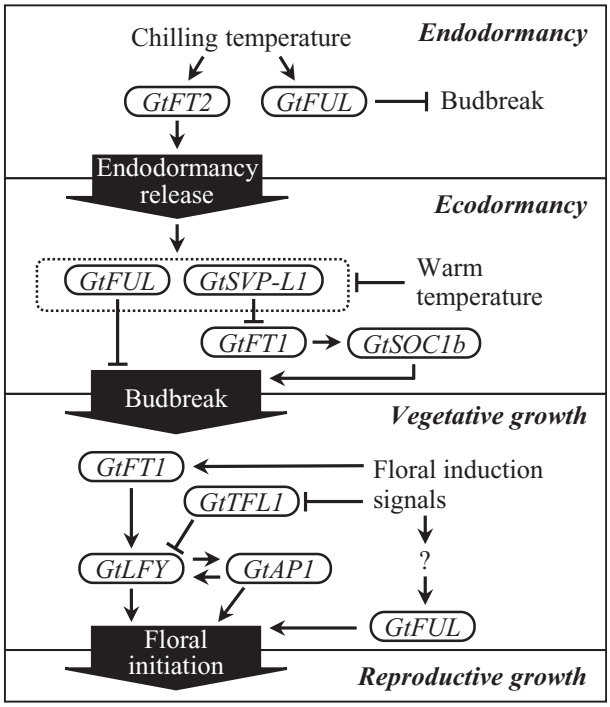


Figure 8 A schematic model for genetic interactions of gentian *FT* orthologs and related genes that regulate developmental phase transitions in gentians.

to ecodormancy. To confirm the role of *GtFUL* expression, we treated the WT OWBs with chilling and found that the expression of *GtFUL* increased and was maintained at a certain level during endodormancy (Supplemental Figure S5), implying that *GtFUL* suppressed budbreak during both endodormancy and ecodormancy phases and that the lack of *GtFUL* leads to extremely early budbreak. With regard to *GtSVP-L1*, previous study revealed that the *SVP* ortholog, *SVP-like* (*SVL*), acts as dormancy promoter and that decreased expression of *SVL* induced early budbreak in hybrid aspen (Singh et al., 2018). On the basis of these findings, we speculated that the expression of *GtFUL* and *GtSVP-L1* was increased to prevent untimely budbreak during dormancy, and decreased when receiving budbreak signals such as warm temperatures (Figure 8). This hypothesis is supported by studies showing that *FUL* and *SVP* also regulate growth arrest and dormancy (Balanzà et al., 2018; Singh et al., 2018). Although *FUL* is known to be involved in several developmental processes, including flowering and fruit development (Gu et al., 1998; Melzer et al., 2008), *FUL* suppresses the expression of *SMALL AUXIN UPREGULATED RNA 10* (*SAUR10*), a member of the *SAUR* family of genes that regulates plant growth through auxin, and promoted the proliferative arrest of the inflorescence meristem to terminate indeterminate flower production in Arabidopsis (Bemer et al., 2017; Balanzà et al., 2018). In contrast, *SVP* promotes dormancy in hybrid aspen buds under ABA-mediated photoperiodic control, and the overexpression of *SVP2* prevents budbreak in kiwifruit (Wu et al., 2017; Singh et al., 2019). Notably, the regulation of budbreak by *SVP* orthologs is accomplished through *FT* orthologs. Singh et al. (2018) demonstrated that the overexpression of an *SVL* gene repressed budbreak in hybrid aspen through the repression of *FT* expression, whereas RNAi-mediated suppression of *SVL* upregulated *FT* expression, resulting in early budbreak. Since *GtFT1* and *GtSVP-L1* showed the opposite expression pattern before budbreak in our study, it is possible that *GtSVP-L1* repressed *GtFT1* expression, and then budbreak was induced. Additionally, because *SVP* represses gibberellin synthesis during Arabidopsis flowering (Andrés et al., 2014), the negative regulation of gibberellins by *SVP* and *SVL* may also contribute to dormancy maintenance in perennial buds. In addition to reports that *FUL* and *SVP* are involved in *FT* and phytohormone regulation, the overexpression of *CsFUL* downregulated cell division and cell cycle-related genes in cucumber (Zhao et al., 2019). *SVL* also affects the cell cycle by regulating *EARLY BUD BREAK 1* and *CYCLIND3.1* genes in hybrid aspen (Azeez et al., 2021). The cell cycle of dormant buds is typically arrested at the G1 phase, and progression from the G1 stage to the S stage is promoted by gibberellin and *CYCLIND3.1* during budbreak (Horvath et al., 2002; Azeez et al., 2021). It is, therefore, possible that *GtFUL* and *GtSVP-L1* repress budbreak through sequential and/or simultaneous regulation of *GtFT1*, the cell cycle, and phytohormones.

Conclusion

In this study, we documented that two gentian *FT* orthologs function at different developmental phases: namely, *GtFT1* functions at the beginning and end of the vegetative growth phase, whereas *GtFT2* functions during the endodormancy phase (Figure 8). Our study supports the notion that *GtFT1* promotes budbreak and floral initiation in concert with other transcription factors. Analysis of *ft2* mutants created by CRISPR/Cas9-mediated genome editing revealed that mutations in *GtFT2* suppressed the transition to the dormancy phase, resulting in a lower budbreak frequency and budbreak delay. This finding indicated that *GtFT2* functions as an accelerator of endodormancy release. The *GtFT2* mutation also suppressed the expression of *GtFUL* and *GtSVP-L1*, two genes considered to repress untimely budbreak during ecodormancy. Our results also highlight the idea that gentian bud dormancy is regulated by signal transmission from one *FT* ortholog to another *FT* ortholog through transcription factors. Furthermore, duplicated *FT* paralogs in gentians may be differentiated by their responsiveness to endogenous and environmental cues and their ability to regulate downstream genes required to set the developmental phases suitable for growth or survival.

Materials and methods

Plant materials

Three-year-old *Gentiana triflora* cv. SpB, Iwate, and *G. scabra* cv. Alta plants grown in an agricultural field in Iwate Prefecture were used to study gene expression in gentians. For seasonal expression analysis of *G. triflora* cv. SpB, OWBs

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were harvested every month from October 2011 to March 2012, and leaves and SAMs of *G. triflora* cv. SpB were harvested every month from April 2011 to September 2012. Five OWBs, leaves, and SAMs were harvested from five different field-grown plants. The leaves and SAMs of the harvested OWBs were separated before analysis. OWBs of *G. triflora* cv. Iwate and *G. scabra* cv. Alta were harvested from field-grown plants every month from October 2010 to April 2011. For the chilling treatment, OWBs of cv. SpB harvested in October 2011 were transferred to 4°C and sampled at 0, 1, 3, 5, and 7 weeks after initiating the treatment. All field samples were harvested at 10:00 to 10:30 a.m. The samples were frozen in liquid nitrogen, lyophilized using a freeze dryer (FDU-2210, EYELA), and stored at −20°C until use. For genome editing, *G. triflora* x *G. scabra* cv. Albireo was used. Plantlets were grown on solid Murashige and Skoog (MS) medium containing 0.2% (w/v) gellan gum at 20 °C with a 16-h/8-h light/dark photoperiod at a light intensity of 40 μmol m^{-2} s^{-1}.

Vector construction and plant transformation

The *ft2* and *ful* mutants were generated by CRISPR/Cas9-mediated genome editing using the gentian cultivar reported by Tasaki et al. (2019). Binary CRISPR/Cas9 vectors were constructed based on the pSbar-pcoCas9 vector with a plant codon-optimized Cas9, SpCas9 (pcoCas9), and a gene for bialaphos-resistance. A sgRNA expression cassette containing a 20-bp nt sequence targeting the *FT2* and *FUL* genes was introduced into pSbar-pcoCas9 as a BstXI–AscI fragment to produce pSbar-pcoCas9-AtU6-26p-FT2-gRNA and pSbar-pcoCas9-AtU6-26p-FUL-gRNA vectors (Supplemental Figure S6) that were then transformed into *Agrobacterium tumefaciens* strain EHA101 by electroporation. Plant transformation was achieved using the leaf disc method optimized for gentian species, as reported by Mishiba et al. (2005). Briefly, leaf sections excised from gentian plantlets (*G. triflora* x *G. scabra* cv. Albireo) were co-cultivated for 5 d with *A. tumefaciens* carrying a binary CRISPR/Cas9 vector. After washing, leaf sections were placed on half-strength solid MS medium containing 0.5 mg L^{-1} bialaphos, 500 mg L^{-1} cefotaxime, 10 mg L^{-1} thidiazuron, and 1 mg L^{-1} 1-naphthaleneacetic acid. Adventitious shoots induced from calli dedifferentiated from leaf sections were transferred to half-strength solid MS medium containing 0.75 mg L^{-1} bialaphos, 500 mg L^{-1} cefotaxime, and 3% (w/v) sucrose for rooting. Transgenic plantlet selections were identified by polymerase chain reaction (PCR) using MightyAmp DNA polymerase (Takara) with specific primers for pcoCas9 (Supplemental Table S1) according to the manufacturer's instructions.

Mutation detection

Genomic DNA was extracted from the leaf tissues of transgenic plantlets using an innuPREP Plant DNA Kit (Analytik Jena) according to the manufacturer's instructions and used as a template. The genomic sequence, including the target sequence, was amplified using PrimeSTAR Max DNA Polymerase (Takara Bio) and the primers shown in Supplemental Table S1. The purified PCR product was cloned using a ZeroBlunt TOPO PCR Cloning Kit (Thermo Fisher Scientific). DNA was sequenced using an ABI PRISM Big Dye Terminator Cycle Sequencing Kit with an ABI 3100 DNA sequencer (Applied Biosystems). A UPGMA dendrogram was generated using CLC Sequence Viewer 8.0 software (CLC Bio). Nucleotides and deduced amino acid sequences were aligned using Genetyx-Mac ver. 19 software (Genetyx Corporation).

Quantification of sugar concentrations

Sugar extraction was performed according to the method of Takahashi et al. (2014). Briefly, lyophilized tissue samples were pulverized in a Micro Smash M100 (TOMY) cell disruptor and homogenized with ice-cold 50% (v/v) methanol containing 10 μM 1,4-piperazineethanesulfonate and methionine sulfone as internal standards for quantification. Homogenates were then centrifuged at 20,000*g* for 5 min, the supernatant liquid was filtered through a 3-kDa-cutoff filter (Millipore) by centrifuging at 16,000*g* for 60 min, and the filtrates were used for analysis. Sugars were quantified by liquid chromatography/quadrupole time-of-flight mass spectrometry (Agilent Technologies) with a Hypercarb column (4.6 × 100 mm, 5 μm; Thermo Scientific; Takahashi et al., 2017). Accuracy was verified with known concentrations of reference standard compounds using Agilent MassHunter Quantitative Analysis Software (Rev. B.05.02).

EST library construction

Normalized cDNA library of *G. triflora* was synthesized from a mixture of total RNA extracted from leaf and SAM of vegetative and reproductive growth phases, OWBs of endodormancy and ecodormancy phases, and flowers of *G. triflora*, sequenced by Roche 454 FLX Titanium instrument and assembled using a GS De Novo Assembler (Newbler) v2.5.3 software (Roche Applied Science). Preparation of the library, sequencing, and sequence assembly was performed as custom service by Generis, Inc (Yokohama, Kanagawa, Japan). Sequences of gentian orthologs were searched by local blast search against the cDNA library using the protein sequences of Arabidopsis or grape FLC, SOC1, FUL, SPL4, AP1, and LFY (Supplemental Figure S3) as queries using Genetyx-Mac version 19 software (Genetyx Corporation).

Molecular cloning and reverse transcription-quantitative PCR analysis

RNA extraction, first-strand cDNA synthesis, and reverse transcription-quantitative polymerase chain reaction (RT-qPCR) analysis were carried out according to the methods described in Takahashi et al. (2014). Briefly, total RNA was extracted using an RNAs-ici!-P kit (Rizo Inc.), and incubated with RNase-free DNase I (Wako) to eliminate genomic DNA. First-strand cDNA was synthesized from 500 ng of total RNA using a TaKaRa RNA PCR kit (AMV) Ver.3.0 (Takara) with an oligo(dT) primer according to the

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manufacturer's instructions. The first-strand cDNA containing a full-length ORF was amplified by PCR, cloned into a pCR-Blunt vector (Thermo Fisher Scientific), and sequenced using an ABI PRISM Big Dye Terminator Cycle Sequencing kit with an ABI 3100 DNA sequencer (Applied Biosystems). For phylogenetic analysis, amino acid sequences were aligned with a gap open cost 10.0 and a gap extension cost 1.0, and phylogenetic tree was constructed by the UPGMA method using CLC Sequence Viewer v8.0 (CLC bio; Supplemental Figure S7). For RT-qPCR analysis, cDNA was synthesized from 500 ng of the total RNA using a High-Capacity cDNA Reverse Transcription kit (Applied Biosystems) according to the manufacturer's instructions and used as the template. RT-qPCR was conducted using a KAPA SYBR FAST ABI Prism qPCR Kit (Kapa Biosystems) on a StepOnePlus Real-Time PCR System (Applied Biosystems). The expression levels of corresponding genes were calculated by the ΔΔCT method (Livak and Schmittgen, 2001). The gentian actin (*ACT*) gene was used as an internal control and to normalize the relative expression levels of all samples. Primers used in this study are listed in Supplemental Table S1.

Statistical methods

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS v.10.0). Statistical analysis was performed with Student's *t* test and a difference at $P < 0.05$ was considered as significant.

Accession numbers

Sequence data from this article can be found in DDBJ databases under the following accession numbers: *GtFT1* (AB605176), *GtFT2* (AB605177), *GtFLC-L* (LC577833), *GtSOC1a* (LC577834), *GtSOC1b* (LC577835), *GtFUL* (LC577836), *GtSPL4* (LC577837), *GtAP1* (LC577838), *GtLFY* (LC577839), *GtSVP-L1* (LC049617), and *ACT* (LC189205).

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Genomic structure and nucleotide sequence of the gentian *FT2* gene.

Supplemental Figure S2. Phenotypes of gentian *ft2* mutants.

Supplemental Figure S3. Alignments of the deduced amino acid sequences of FLC, SOC1, FUL, SPL4, AP1, and LFY in gentian, Arabidopsis, and grape.

Supplemental Figure S4. CRISPR/Cas9-mediated mutations in the gentian *FUL* gene.

Supplemental Figure S5. Effect of chilling on the expression of *GtFUL* in WT overwintering buds.

Supplemental Figure S6. Schematic diagrams of the T-DNA region of the binary vectors used for genome editing in this study.

Supplemental Figure S7. Sequence alignment and phylogenetic tree based on the deduced amino acid sequences of GtFUL and its homologs from *Arabidopsis thaliana* (AtFUL), *Cornus kousa* (CkFUL1), *Hydrangea macrophylla* (HmFUL1), and *Vitis vinifera* (VvFUL) using the UPGMA method.

Supplemental Table S1. A list of gene accession numbers and primers used in this study.

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Conflict of interest statement. None declared.

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